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3-12-2024

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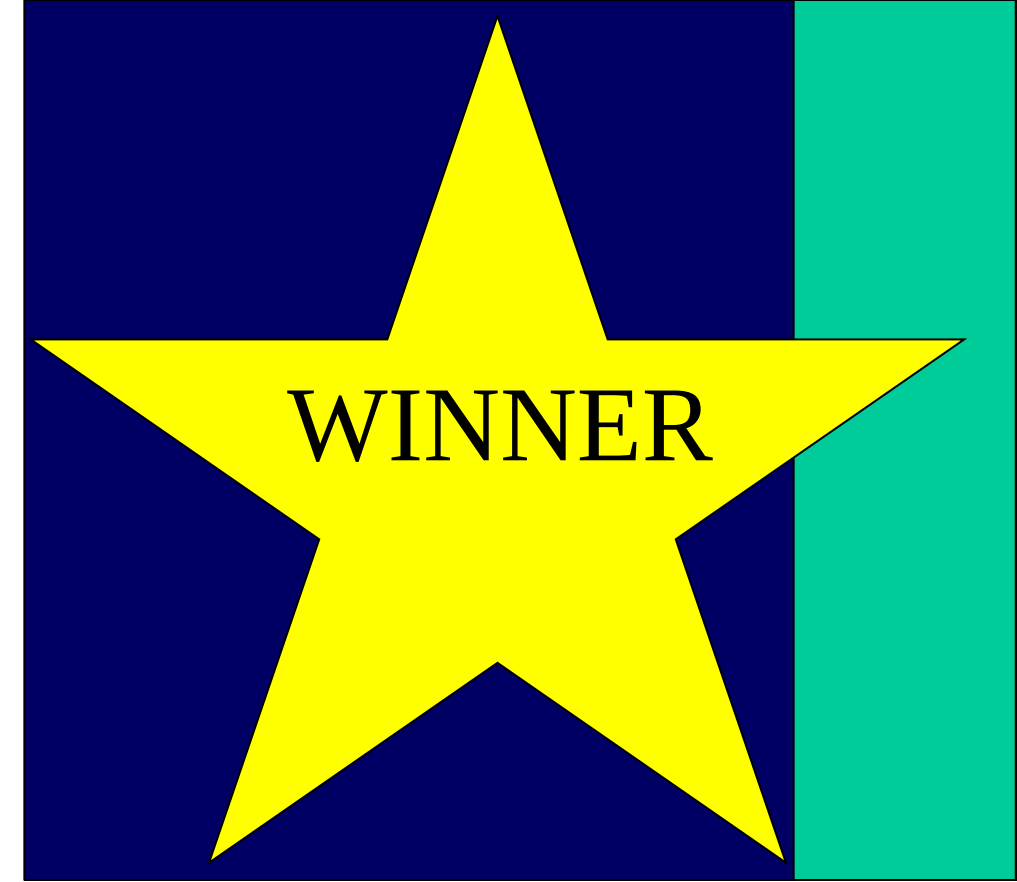
Recommended Citation

Plemmons, Elizabeth; Ribitch, Trisha; Kaps, Zoe; and Engelbrecht, Asma, "Project #30: Reduction in use of CXR for Children with Asthma, Bronchiolitis, and Croup" (2024). *Quality Expo 2024*. 12.
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Reduction in use of CXR for Children with Asthma, Bronchiolitis, and Croup

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AIM

Elements to Consider:

- Problem Statement:** Respiratory illnesses particularly asthma, bronchiolitis and croup are among the top diagnoses for which children visit the emergency department. Chest x-rays are frequently used by emergency providers in children presenting with these conditions, yet they rarely reveal information that contribute to changes in management and can lead to excess cost, radiation exposure, & overtreatment with antibiotics.
- Improvement (Goal) Statement:** Reduce the use of chest x-rays in children who present to the ED with croup, asthma, and bronchiolitis to 20% or less by the end of 2023.

BASELINE DATA

Elements to Consider:

- Base line data**
Average is 55% in 2021. A goal in Reduction of chest Xray procedures to lessen radiation for Pediatric patients. All ED providers are engaged in continued education to improve quality of care in pediatric patients.
- Plan-** Educate physicians monthly in the emergency department on the importance of utilizing tools that were developed to aid in decision making regarding chest x-rays. Track data on chest x-ray appropriateness per provider. Provide targeted education to continue to improve quality of care.

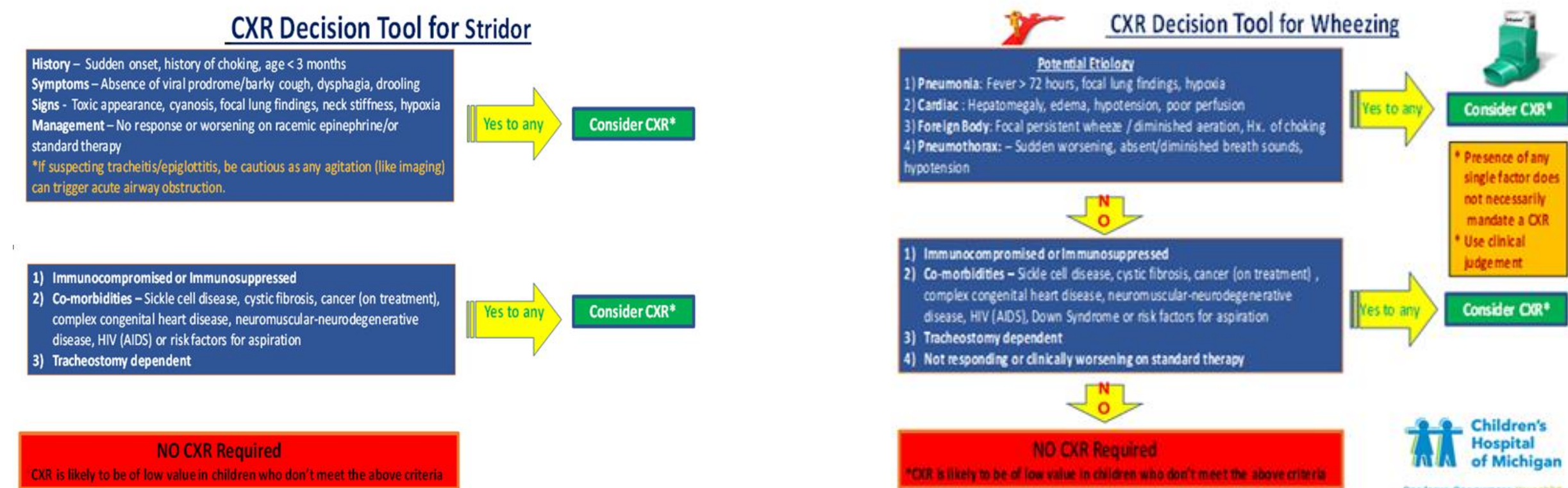
INTERVENTIONS

- Low value Respiratory CXR guidelines provided by MEDIC to guide providers in making clinical decision-making regarding chest x-rays
- Monthly review of the guidelines with providers throughout 2022.
- APP teaching took place in early 2022 to introduce guidelines for pediatric chest x-rays, as well as completing in depth chart reviews.

If NONE of the below are present, question your reason for a CXR.			
	ASTHMA 2-17 yrs old	BRONCHIOLITIS 2mo-2yrs old	CROUP 6mo-3yrs old
HISTORY OF PRESENT ILLNESS	Fever ≥38°C (100°F) for ≥72 hrs		
	Chest pain		
	Suspected foreign body ingestion or choking episode in past 2 wks		
PAST MEDICAL HISTORY	• Cerebral palsy &/or neuromuscular disease • Prematurity (<37 weeks gestation) • Bronchopulmonary dysplasia • Tracheostomy • Immunosuppression (cancer, HIV/AIDS, transplant)		• Cystic fibrosis • Ciliary dyskinesias • Congenital heart disease • Sickle cell disease
EXAM FINDINGS	Toxic, ill appearance, somnolent, lethargic, or listless		
	Focal lung exam findings (decreased breath sounds, rales, rhonchi) or crepitus		
CLINICAL COURSE	Worsening clinical status: Vital signs and/or exam findings and/or requiring escalation of care		
Presence of one or more of these does NOT automatically require a CXR.			
*If wheezing is occurring without a clear atopic etiology or URI symptoms, diagnostic imaging may be considered on a case-by-case basis.			

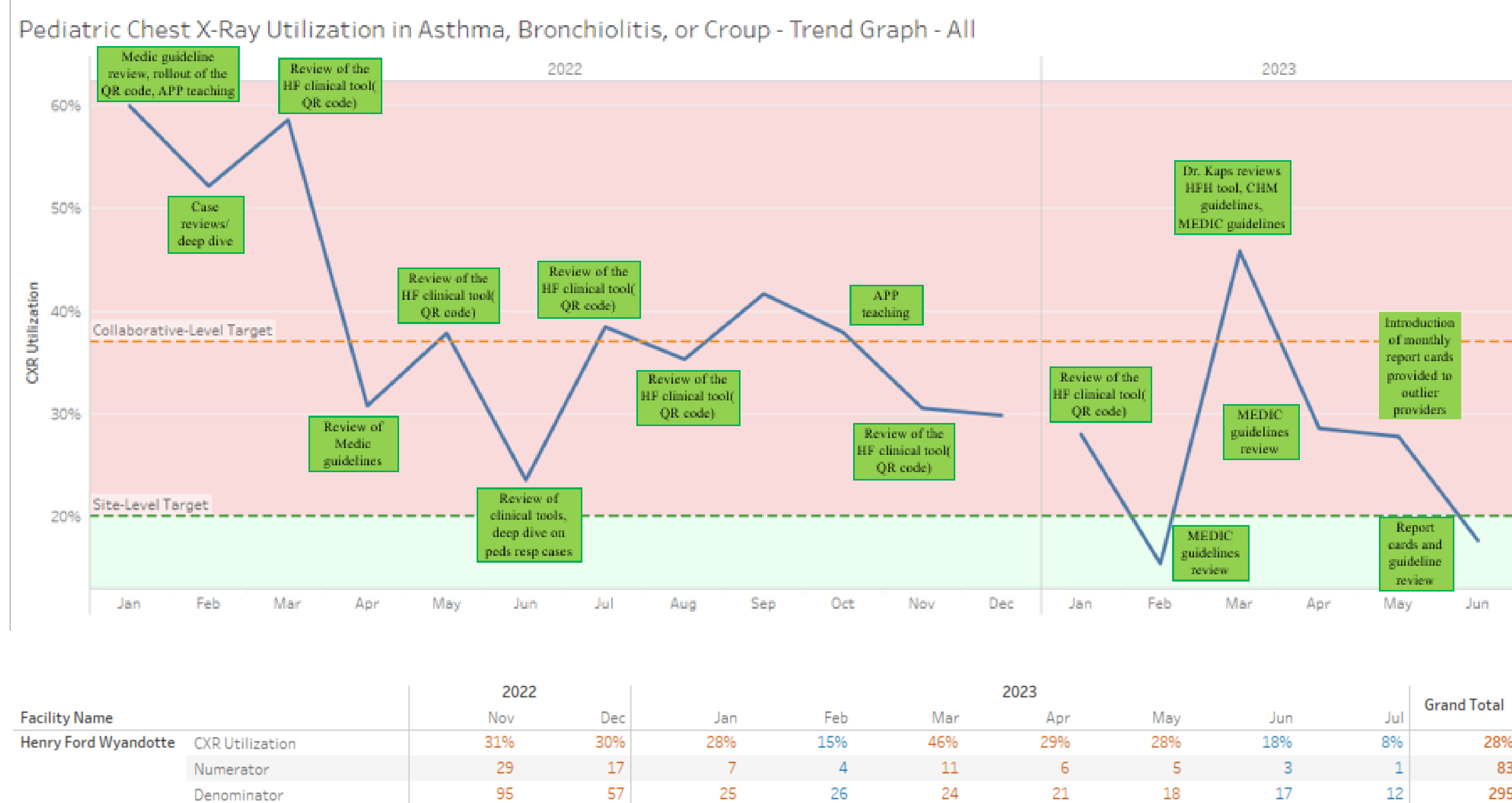
INTERVENTIONS CONT.

- January 2022 a QR code was developed for use by HFWH ED providers to aid in decision making regarding pediatric chest x-ray. This QR code is available to providers to pull up when they are treating pediatric patients. The QR code is intended to act as a map to aid in the decision making if a patient requires a chest x-ray or not.
- June 2022 review of pediatric chest x-ray cases with providers to give case examples.
- Additional teaching was provided to the APP's at the end of 2022.
- March 2023 a detailed presentation was given to the ED providers by Dr. Kaps, MD a senior resident. This included review of MEDIC guidelines, CHM tool for chest x-rays, and the QR code is updated to the CHM tool.



- July 2023 targeted outlier providers (bottom 5) and provided "report cards" which detailed cases that did not show appropriate use of chest x-ray.
- Monthly "report cards" will continue to go out to outlier providers. Individualized educational materials will also be sent with the report card to drive change and quality improvement.

EVALUATION OF CHANGES



Timeline of interventions in correlation with the utilization of chest x-rays on pediatric patients presenting with croup, bronchiolitis, and asthma.

EVALUATION OF CHANGES CONT.

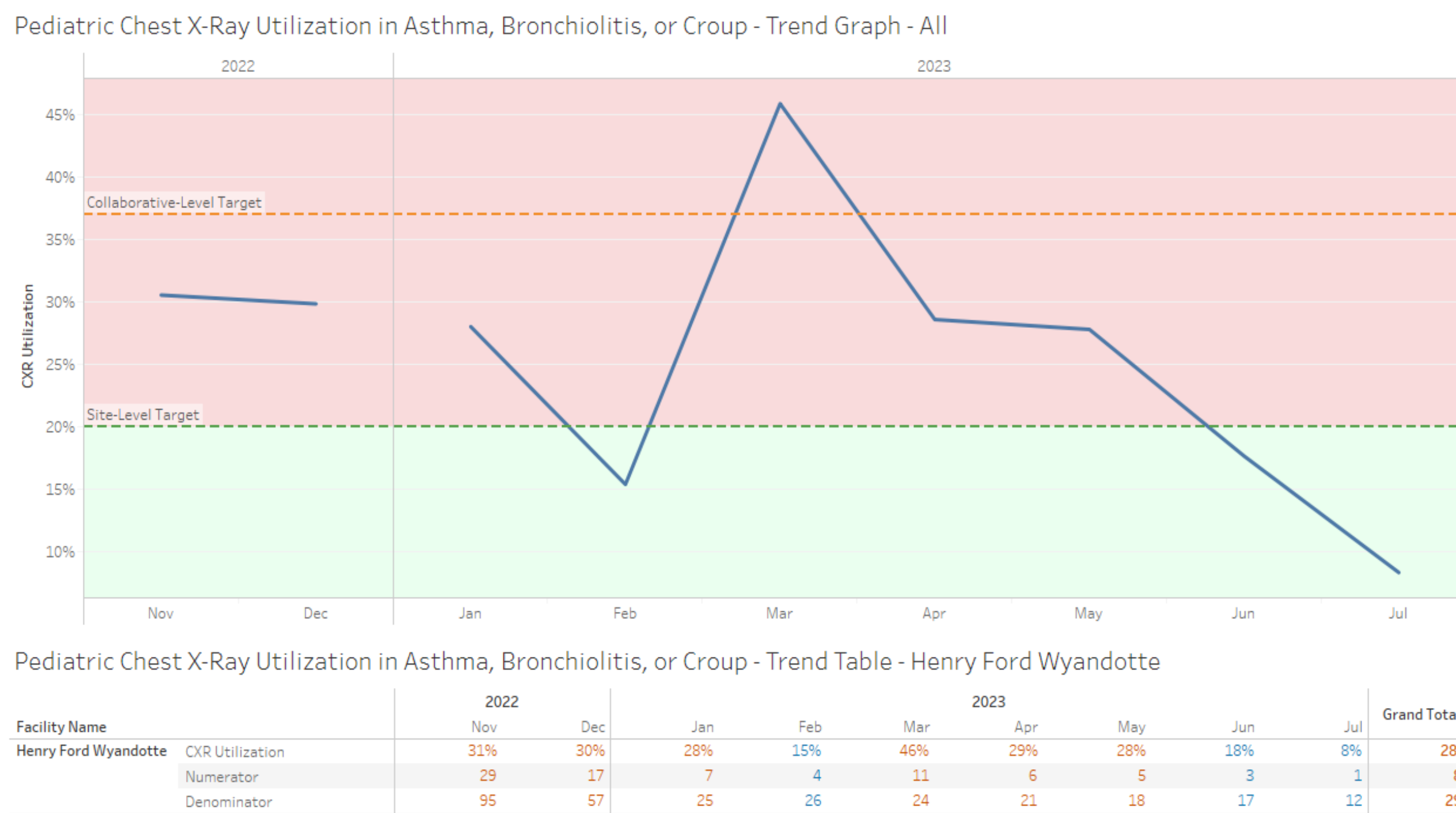
Year	Highest Month	Lowest Month	End of Year Average
2021	83%	23%	62%
2022	63%	24%	44%
2023	46%	8%	28%

Year over year overall reduction in chest x-rays for pediatric patients presenting in the ED

- Monthly evaluation of data was collected through MEDIC abstraction through in-depth chart review.
- Monthly analysis of data was used to design educational materials, and interventions to improve quality of care for pediatric patients presenting with croup, bronchiolitis, and asthma.
- Analysis of results is ongoing and will continue throughout 2023 to continually improve care across the continuum for our pediatric patients. This as well will improve cost savings and improve safety.

ACT: SUSTAIN AND SPREAD

- Ongoing data post implementation plan-** Continuation of measuring the utilization of chest x-ray use in pediatric patients presenting to the ED with croup, bronchiolitis, and asthma.
 - Data will be broken down by provider.
 - Each outlier case will have an extensive chart review.
 - Results of the chart review will be emailed to the providers monthly.
 - Monthly education will also be provided to help to further educate providers.
 - Continue to emphasize the use of the QR code to the providers before ordering a chest x-ray for a pediatric chest x-ray with a specific respiratory complaint.
- 2023 goal-** Currently the goal for 2023 is 20% or less. Our current data is at 28% and improving monthly with continuation of multiple intervention implementation.



- Sustainability-** The team intends to continue the interventions to improve care for pediatric patients regarding minimizing the use of chest x-rays in pediatric patients.
- Continuing "report cards" for providers.
- One on one meetings with providers who struggle meeting goals in reducing their use of chest x-ray.
- Ongoing monthly education during department meetings to emphasize the metric to the physicians and APPs